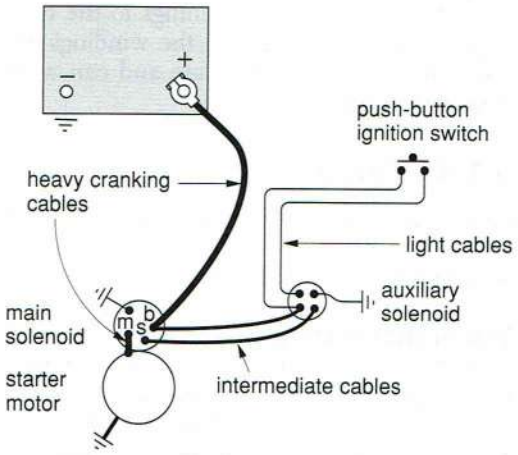


FIGURE 7-22C. A starting circuit with main and auxiliary solenoids. To bypass the switch circuit, connect a jumper from B to S on the main solenoid. To bypass the solenoid and switch, connect a heavy-duty jumper from B to M.

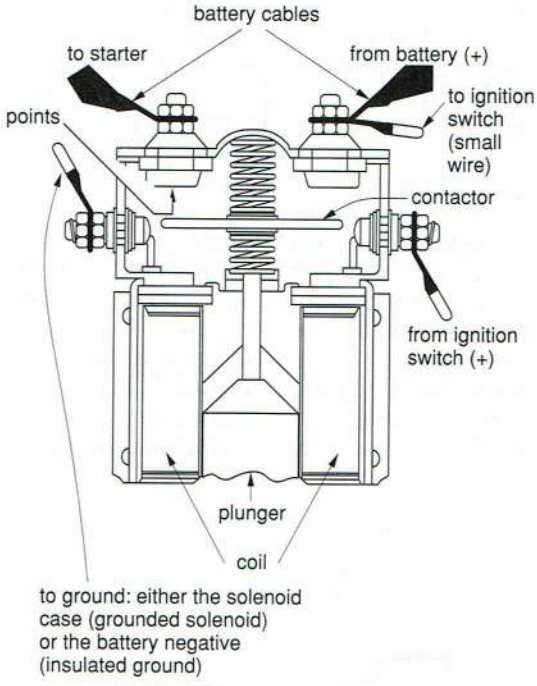


FIGURE 7-22D. Solenoid operation. Upon closing the ignition switch, a small current passing through the solenoid's electromagnet pushes the contactor up against spring tension, completing the circuit from the battery to the starter motor.

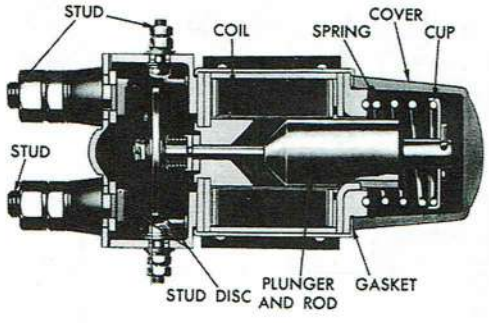


FIGURE 7-22E. A cutaway of a typical solenoid. *(Detroit Diesel)*

cranked when it is in gear. Some heavy-duty circuits have two solenoids. The first is energized by the ignition switch to close the circuit to the second, which then operates the starter motor. (This is done because the main solenoid itself draws a moderately heavy current, requiring fairly heavy cables. Adding a second light-duty solenoid enables lighter wires to be run to the ignition switch.)

The ground side of a starter motor circuit almost always runs through the starter motor case and engine block to a ground strap, which is connected directly to the battery's negative terminal post. However, on rare occasions an insulated ground is used, in which case the starter motor is isolated electrically from the engine block; a separate cable grounds it to the battery.

Inertia and preengaged starters. Most starter motors are series-wound motors although some newer ones have permanent magnets. Regardless of motor type, there are two basic kinds of starters: inertia and preengaged. Inertia starters are mostly found on older engines; as far as I know, all modern engines have preengaged starters. Some newer starter motors have a reduction gear (generally between 2:1 and 3.5:1) between the electric motor and the drive gear (the *pinion* gear) that engages the engine flywheel. This enables a smaller, lighter, faster-spinning motor to do the work of a larger motor.

Inertia starters. The solenoid for an inertia motor is mounted independently, at a convenient location. The pinion gear is keyed into a helical groove on the motor shaft. When the solenoid is energized, the motor spins. Inertia in the gear causes it to spin out along the helical groove and into contact with the engine flywheel, which is then turned over (Figure 7-23A).

Sometimes an inertia starter's pinion will stick to its shaft and not be thrown into engagement with the flywheel. In this case, the motor will whir loudly without turning the engine over. A smart tap on its case while it is spinning may free it up. However, don't do this too often, as it's hard on the gear teeth. Also, don't do this to a permanent-magnet motor—it may crack the magnets or break them loose from the motor case (as far as I know, there are no inertia starters with permanent-magnet motors).

At other times the pinion may jam in the flywheel and not disengage. In this case you can often gain access to the end of the armature shaft via a cover, normally located in the center of the rear housing of the starter motor. *With the engine shut down*, turn the squared-off end of the shaft back and forth with a wrench to free the pinion.

Preengaged starters. The solenoid of a preengaged starter is *always* mounted on the starter itself (a surefire indicator). When the solenoid is energized, the electromagnet pulls a lever, which pushes the starter motor pinion into engagement with the engine flywheel; the main solenoid points now close, allowing current to